

SHIVAM MAURYA

AI/ML Engineer

Pune, India | [Phone](#) | [Email](#) | [Github](#) | [Linkedin](#)

TECHNICAL SKILLS

Languages: Python (primary) · C++ · SQL · Bash

ML / Deep Learning: TensorFlow · Keras · PyTorch · Scikit-Learn · XGBoost · LightGBM · CNN · Transfer Learning (Xception, ResNet) · Model Evaluation & Hyperparameter Tuning

Generative AI / LLMs: LangChain · LangGraph · Multi-Agent Orchestration · Prompt Engineering · Retrieval-Augmented Generation (RAG)

Computer Vision: OpenCV · YOLO · Image Classification · Object Detection · Pose Estimation · ONNX Runtime · Quantization

MLOps / Deployment: Docker · Git/GitHub · REST APIs · FastAPI · Streamlit Cloud · CI/CD · Postman · Jupyter · Google Colab · Linux (Ubuntu)

CS Fundamentals: Data Structures & Algorithms · Object-Oriented Programming · Database Management Systems (SQL) · Version Control

EXPERIENCE

AI/ML Systems Developer — Student Researcher

Jun 2024 – Present

Army Institute of Technology, Pune

- Designed and deployed modular **AI inference pipelines** integrating YOLO object detection and CNN classifiers, achieving **30+ FPS real-time inference** on CPU-only embedded hardware without GPU acceleration.
- Exported trained **TensorFlow** models to **ONNX** for runtime-agnostic, optimized inference; implemented hot-swappable model backends enabling A/B evaluation of model versions in production.
- Built computer vision perception systems with **OpenCV** for real-time classification, object detection, and pose estimation, integrating outputs into downstream decision pipelines.
- Trained **CNN and Xception transfer-learning models** for multi-class classification; applied data augmentation, cross-validation, and hyperparameter tuning to reach production-grade accuracy.

Open-Source Contributor — Robotics & Systems Software (ROS 2 Ecosystem)

Jun 2025 – Present

navigation2, ros2_control, ros2_controllers, ros2/demos, ros2_documentation

- Shipped 15+ merged pull requests** and maintain 5+ open PRs across large-scale, multi-maintainer open-source C++/Python repositories, collaborating with global engineering teams via Git-based code review and CI pipelines.
- Diagnosed and fixed CI/build regressions (including a cross-project initialization-strictness bug), improved developer documentation, and added static-analysis linting (ament_mypy) to production codebases.

Software Lead (previously Joint Secretary) — Centre of Excellence for AI & Robotics

Jun 2025 – Present

Army Institute of Technology, Pune

- Lead technical direction for a **100+ member research cell**; designed and deployed **4+ end-to-end ML pipelines** (CNN, Xception, Random Forest, XGBoost) with data augmentation and hyperparameter tuning.
- Coordinate 5+ cross-functional student research teams on AI perception and intelligent systems, reporting technical progress to faculty leadership.

PROJECTS

Credi-Mitra — Agentic AI Credit Appraisal System

Python, XGBoost, LangGraph, LangChain, Streamlit

- Built a **multi-agent, LangGraph-orchestrated pipeline** that autonomously evaluates loan applications with **97% accuracy**, minimizing manual underwriting steps.
- Designed an agentic decision flow where the agent selects an evaluation strategy, retrieves supporting context, and generates human-readable appraisal reports; **deployed live on Streamlit Cloud** — Runner-up, Hackathon 2026.

MedSynapse — Multi-Disease Clinical AI Platform

Python, TensorFlow, Xception, ONNX, Keras, Streamlit

- Trained an **Xception transfer-learning model** for 4-class brain MRI tumor classification (299×299 input) with data augmentation and stratified cross-validation, achieving production-grade diagnostic accuracy.
- Exported to **ONNX** for runtime-agnostic deployment as 1 of 4 integrated ML/DL diagnostic modules; selected for the Diagnostic Tools track, IIIT Pune Thinking Machines 2025.

On-Device AI Inference Engine

Python, C++, TensorFlow, ONNX Runtime, OpenCV, YOLO

- Built a production-style inference pipeline with hot-swappable YOLO/CNN backends, achieving **30+ FPS on CPU-only hardware** via ONNX graph optimization and quantization.

- Designed a modular, JSON-based architecture enabling model swaps with **zero downstream code changes**, mirroring MLOps best practices; validated live under real-time competition conditions (ABU Robocon 2025/26).

Antimicrobial Resistance (AMR) Prediction

Python, Scikit-Learn, Random Forest, XGBoost, Stratified K-Fold CV

- Predicted antibiotic resistance across 274 bacterial isolates, handling 99.3% class imbalance with class-weight balancing and stratified 5-fold cross-validation.
- Best model (Random Forest) achieved **F1 = 0.7982, accuracy = 99.27%** across 7 evaluated classifiers, with decision thresholds aligned to WHO/CLSI clinical standards — **top submission, CODECURE Hackathon 2025**.

AI Predictive Maintenance System

Python, Scikit-Learn, Random Forest, Feature Engineering, Streamlit

- Trained a Random Forest classifier (**98%+ accuracy**) on the AIAI 2020 industrial sensor dataset to predict equipment failures from temperature, torque, RPM, and tool-wear signals.
- Built a dual-mode Streamlit dashboard — Manual Diagnostics for single-unit inspection and Fleet Monitoring with live failure-probability indicators across **9+ machines**.

EDUCATION

B.E. Automation and Robotics Engineering — Army Institute of Technology, Pune

Jun 2024 – May 2028

CGPA: 8.0 / 10

Relevant Coursework: Machine Learning · Computer Vision · Data Structures & Algorithms · Database Management Systems (SQL) · Embedded AI Systems · Control Theory · Probability & Statistics

ACHIEVEMENTS & EXTRACURRICULARS

- **IIIT Pune Thinking Machines 2025** — MedSynapse platform selected for the Diagnostic Tools track (4 integrated ML/DL models).
- **CODECURE Hackathon 2025 (Track B), IIT BHU** — Top submission; AMR prediction system at 99.27% accuracy, WHO/CLSI-aligned.
- **Open-Source Contributor** — 15+ merged pull requests, 5+ open, across large-scale robotics/systems repositories (navigation2, ros2_control, ros2_controllers).
- **ABU Robocon 2025/26** — Led software architecture and real-time CNN-based vision deployment on the competition robot; represented AIT Centre of Excellence for AI and Robotics.